Source, capacity and the selection of field energy

From complete sources to spherical motion

Erik Echeverria
Rough Draft 2 | 1 November 2026

Matter continuously commits field interaction capacity as gravitational burden. SET expresses this commitment through its source law and relates the remaining capacity to translation and local physical evolution. The question developed here is how those source and capacity requirements restrict the energy of the field itself.

I begin with independent clock and spatial responses and unknown local field-energy coefficients. The same properly measured weak-source law is required in loaded surroundings, and the actual source law is applied to complete closed sources, including their field and physical supports. Supported hollow sources determine the exterior response; interacting sources determine the remaining energy coefficients on the connected domain they probe. Clock homogeneity follows from their force and energy identities. The selected energy supplies the work current, the spherical clock and radial response, and, through the established probe coupling, the perihelion equation.

The main theorem concerns a regular local static two-response class and an attached weak source-covered domain. A conditional continuation along the regular finite-load source family is developed separately in Appendix D. The mathematical steps and the source preparations that justify them are kept together throughout the derivation.

1. Capacity: one physical limit for motion and events

In SET, a field's limited capacity for interaction means a limited capacity for physical motion. Mass continuously commits part of that capacity as gravitational burden, with its sourcing quantified by the law Q. Greater burden leaves less capacity available for translation and the local motion through which events unfold. Greater translational speed takes a larger share of what remains, leaving less for internal progression per reference second. A clock's journey and its ticks participate in the same physical limit.

$$S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2, \qquad \beta = \frac{d\tau}{dT}. \tag{1}$$

Here T is the specified reference time and τ is the probe's proper time. The burden S and translational reading V_{rel} have speed units. For a held clock, translation vanishes and its progression is α. Define its square A:

$$\alpha^2 = A = 1 - \frac{S^2}{c^2}, \qquad \beta^2 = A - \frac{V_{\mathrm{rel}}^2}{c^2}. \tag{2}$$

The local observer uses clocks and rulers participating in the same field response. The locally measured limiting speed is c; propagation measured against the reference time depends on that response. The radial propagation law will be calculated after the spatial response has been selected.

The capacity relation also has an exact quadratic energy reading. Multiplying every term by m/2 gives

$$K_0 = \frac{mV_{\mathrm{rel}}^2}{2} = \frac{mc^2}{2}(1-\beta^2) - \frac{mS^2}{2}, \qquad dK_0 = -\frac{m}{2}\,dS^2 - \frac{mc^2}{2}\,d\beta^2. \tag{3}$$

Motion and clock progression cannot be varied independently of burden. The complete energy framework used below determines how this balance is realized by actual sources and probes. The stationary sources are equilibria of that same energy, so their supporting stresses belong in the calculation.

2. The source law and the energy of a complete source

For a specified spherical source, the source law can be written in its surface form or its volume form:

$$Q_s = 4\pi\sqrt{2GMR_0^3}, \qquad Q_s^2 = 24\pi GMV_{\mathrm{ref}}, \qquad M = \frac{E_s}{c^2}, \qquad V_{\mathrm{ref}} = \frac{4\pi R_0^3}{3}. \tag{4}$$

For later motion calculations define the source strength μ_Q, with units of $\mathrm{m}^3\,\mathrm{s}^{-2}$:

$$\mu_Q \equiv \frac{Q_s^2}{24\pi V_{\mathrm{ref}}} = \frac{GE_s}{c^2} = GM. \tag{5}$$

R_0 is the actual outer areal radius: its surface has physical area $4\pi R_0^2$. The complete isolated energy E_s includes material, binding, field energy and every physical support. It includes the exterior field tail as well as the material inside the boundary. The source-law volume V_{ref} is specified by R_0; the physical volume integral will be defined by the spatial response.

At this actual source surface, the area reading of the law is

$$Q_s = 4\pi R_0^2 S_0, \qquad S_0 = S(R_0). \tag{6}$$

Squaring this reading and comparing it with the source law gives the source boundary condition:

$$S_0^2 = \frac{2GE_s}{c^2R_0}, \qquad \alpha_0^2 = 1 - \frac{2GE_s}{c^4R_0}. \tag{7}$$

The same law is applied to closed sources assembled from ordinary material, including supported hollow sources and a source surrounded by another physical layer. Each

assembly has its own complete energy and actual outer boundary. The field and support energies are evaluated for that assembly.

A distant or weak added layer provides a controlled comparison between source states. The admitted family has regular cluster limits: its small interactions and induced readjustments tend to zero as the separation grows. This source-composition requirement is used explicitly when complete energy is related to the distant clock response.

Reading	Meaning	Units / role
$S,\ V_{\mathrm{rel}}$	Burden and translational motion readings	$\mathrm{m\,s^{-1}}$
$\alpha,\ \beta$	Held and moving clock progression per reference time	Dimensionless
Q_s	Source rate for its specified complete domain	$\mathrm{m^3\,s^{-1}}$
$E_s,\ M$	Complete source energy and its mass equivalent	$\mathrm{J},\ \mathrm{kg}$
$R_0,\ r,\ \rho$	Actual source areal radius, evaluation areal radius, isotropic reference radius	m
$V_{\mathrm{ref}},\ dV$	Specified source-law volume; physical volume element	$\mathrm{m^3}$

3. The same local source law in a loaded field

Consider a small experiment in an approximately uniform background burden S_b, with held-clock factor $\alpha_b > 0$. Its background proper time is $d\tau_b = \alpha_b\, dT$. Subtract the background commitment from the capacity equation and divide by the remaining background capacity:

$$(S^2 - S_b^2) + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2 - S_b^2 = c^2\alpha_b^2. \tag{8}$$

Define the added burden and the local motion and clock readings by

$$S_{\mathrm{local}}^2 = \frac{S^2 - S_b^2}{\alpha_b^2}, \qquad V_{\mathrm{local}} = \frac{V_{\mathrm{rel}}}{\alpha_b}, \qquad \beta_{\mathrm{local}} = \frac{\beta}{\alpha_b},$$
$$S_{\mathrm{local}}^2 + V_{\mathrm{local}}^2 + c^2\beta_{\mathrm{local}}^2 = c^2. \tag{9}$$

The normalization preserves the capacity relation exactly. The physical requirement applied to this local experiment is:

> **Local source universality.** A weak added source obeys the same calibrated source law when its energy, radius and added burden are read using the physical clocks and rulers of its approximately uniform surroundings.

At leading weak-source order, for proper source mass m and proper local radius r_{local}, this means

$$S^2_{\mathrm{local}} = \frac{2Gm}{r_{\mathrm{local}}} + O\left(\frac{G^2m^2}{r^2_{\mathrm{local}}c^2}\right). \tag{10}$$

Equation (9) derives the local normalization. The requirement in (10) states that the actual added source has the same properly measured response. Converting a rate to Q/α_b is a change of clock reading; the local-source requirement additionally constrains the field's response to that source.

This requirement is imposed on the regular ambient-response domain used by the local field equations. It applies to neighboring physically compared background states, so the derivatives of the response are part of the same statement. The global source boundary continues to use the complete energy of the whole assembly.

4. Leave the field-energy coefficients unknown

Use two independent operational responses. The spatial reading is conformal to the declared Euclidean reference ruler, and the second response combines the clock and spatial factors:

$$q_{ij} = \psi^4\delta_{ij}, \qquad dV = \psi^6 d^3x, \qquad \chi = \alpha\psi, \qquad u = (\psi, \chi). \tag{11}$$

ψ and χ are dimensionless. Their relationship is initially unsolved. In spherical symmetry the areal radius is $r = \rho\psi^2$. Gradients written ∇ use the reference ruler; D denotes differentiation in the physical spatial reading q_{ij}.

Examine the regular local quadratic first-gradient energy for these responses:

$$\begin{gathered} E = E_m + F, \qquad E_m = \int \alpha\varepsilon_m\, dV, \qquad \kappa = \frac{c^4}{2\pi G}, \\ F = \frac{\kappa}{2}\int \left[H_{11}(u)|\nabla\psi|^2 + 2H_{12}(u)\nabla\psi\cdot\nabla\chi + H_{22}(u)|\nabla\chi|^2\right] d^3x. \end{gathered} \tag{12}$$

The symmetric coefficient matrix $H(u)$ is unknown, regular and nondegenerate. It is a field-energy coefficient matrix, distinct from a probe Hamiltonian or a Hubble rate. The constant κ has force units, so the integral has energy units. The physical energy density ε_m includes material and support, and the same material energy determines its stress.

This class specifies two response variables, a local first-gradient energy, a regular unloaded state and a fixed boundary-energy convention. Regularity means twice continuously differentiable coefficients in the operational variables near the unloaded state. Source fields approach the reference state at infinity. The same stationary energy supplies both field variations and the mechanical force balance of the supports.

For a test body, the retained common-energy coupling generates translational motion and internal clock progression. Its canonical, test-body homogeneity and clock conditions give

$$E_p = \alpha\sqrt{E_0^2 + c^2 q^{ij} p_i p_j}, \qquad \frac{\partial E_p}{\partial E_0} = \beta, \qquad E_0 = mc^2. \tag{13}$$

Here p is material canonical momentum. Appendix E spells out this retained coupling. The field-space momenta used in the source calculations are different objects and carry their normalization explicitly.

The local added-source experiment constrains the unknown matrix. At a uniform ambient state define $e_\alpha = (-\alpha, 1)$. Linearizing the same energy in a small proper source gives

$$\frac{G_{\text{local}}}{G} = \frac{e_\alpha^T H^{-1} e_\alpha}{2\alpha} = 1. \tag{14}$$

The factors follow by adding a proper point energy $E_{\text{local}} = mc^2$ to a uniform background. Its linear material covector is $E_{\text{local}} e_b/\psi_b$, where $e_b = (-\alpha_b, 1)$ and H_b is the background matrix. The Green function of the reference Laplacian gives

$$\begin{gathered}\delta u = -\frac{E_{\text{local}}}{4\pi\kappa\psi_b\rho} H_b^{-1} e_b, \qquad r_{\text{local}} = \psi_b^2 \rho, \\ \frac{\delta\alpha}{\alpha_b} = -\frac{Gm}{c^2 r_{\text{local}}} \frac{e_b^T H_b^{-1} e_b}{2\alpha_b}.\end{gathered} \tag{15}$$

The coefficient of the proper local potential is precisely the ratio in (14). The same local source law therefore restricts the energy coefficients before an exterior or orbit is prescribed.

The remaining derivation tests these coefficients against actual complete sources. The positive-clock, ordinary attractive branch and the regular source preparations are maintained throughout. The main selection result is exact on the connected weak domain reached by those preparations; Appendix D states the finite-load continuation conditions used to extend the application.

5. Actual sources select the exterior response

The source law constrains the complete physical preparation: its material, supporting motion, binding and entire field tail. Begin by keeping its complete energy E_s distinct from the charge E_C inferred by a distant clock. With ρ the isotropic reference radius and $r = \rho\psi^2$ the areal radius, define

$$\begin{gathered}\psi = 1 + \frac{a_\psi}{\rho} + O(\rho^{-2}), \qquad \chi = 1 + \frac{a_\chi}{\rho} + O(\rho^{-2}), \\ \mu = a_\psi - a_\chi, \qquad E_C = \frac{c^4\mu}{G}, \qquad \alpha^2 = 1 - \frac{2GE_C}{c^4 r} + O(r^{-2}).\end{gathered} \tag{16}$$

Enclose an admitted source A in a real supported shell whose areal radius L tends to infinity and whose proper energy tends to m_B. Regular source composition gives the following limits. The shell support, interaction energy and reaction of the inner source are included in the remainders:

$$E_{AB} = E_A + m_B + o(1), \qquad E_{C,AB} = E_{C,A} + m_B + o(1),$$
$$E_{AB} - E_{C,AB} = o(1) \quad \Longrightarrow \quad E_A = E_{C,A}. \tag{17}$$

The second line follows by applying the exact source law at the aggregate's actual outer material boundary. It earns energy-charge agreement for every admitted inner source. Iterated enclosure permits the same argument for an aggregate itself. Appendix B gives the matching details; no vacuum surface is assigned the status of a material source.

For each exterior use its own charge to normalize the inward radial parameter. The vacuum field equation is a geodesic equation in the two response variables, with constant quadratic norm:

$$y = \tfrac{\mu}{2\rho}, \qquad e = (1, -1), \qquad p^{(y)} = Hu_y, \qquad K_y = u_y^T H u_y,$$
$$\tfrac{dp_a^{(y)}}{dy} = \tfrac{1}{2} u_y^T (\partial_a H) u_y, \qquad \tfrac{dK_y}{dy} = 0, \qquad e^T u_y(0) = 2. \tag{18}$$

Here $p^{(y)}$ is response-space momentum, not particle momentum; K_y is a signed quadratic norm. Actual weak hollow shells exist before their exterior is selected. Their own energy, traction and second-order source law imply $H_\infty e = e$, where $H_\infty = H(1, 1)$ (Appendix A). Their initial momenta are allowed to vary during that argument.

The retained proper local source response, (14), then supplies an asymptotic test for any exterior. Write $\mathcal{D} = a_\psi + a_\chi$. Enclose the same source using either of two distinct finite limiting shell energies. Exact source matching gives a quadratic condition with at most one nonzero root:

$$j_B + \mathcal{C}_\infty j_B^2 = 0, \qquad j_B = \tfrac{2\mathcal{D}}{\mu + G m_B / c^4},$$
$$m_{B,1} \neq m_{B,2} \quad \Longrightarrow \quad \mathcal{D} = 0, \qquad u_y(0) = e, \quad p^{(y)}(0) = e, \quad K_y = 2. \tag{19}$$

The coefficient $\mathcal{C}_\infty$ belongs to the candidate energy, independently of the enclosing mass; Appendix B derives it. Two different masses cannot supply two different nonzero roots. Thus every passing source starts the same normalized vacuum geodesic. Ordinary uniqueness for the regular field equations makes that exterior curve common to the admitted source family.

An actual hollow shell supplies values along this curve. Let U be its proper material energy, P its proper area-integrated tangential pressure, $E_m = \alpha U$, and $P_g = \alpha P$. The jumps, independent radius balance and complete energy give

$$\begin{gathered} \chi p_\chi^{(y)} = -\tfrac{E_m}{E_s}, \qquad \psi p_\psi^{(y)} = \tfrac{E_m + 4P_g}{E_s}, \qquad F_{\mathrm{ext}} = 2P_g = E_s y, \\ \psi p_\psi^{(y)} = 1 + y, \qquad \chi p_\chi^{(y)} = y - 1, \qquad \psi^2 - \chi^2 = 4y. \end{gathered} \tag{20}$$

The pressure is supplied by the shell's own moving particles and is already in U. The last equality is the source law at that shell, using $r = \rho\psi^2$. Varying actual weak shells supplies an open endpoint interval, so these endpoint identities hold along the common curve. Define $L_u = \ln(\psi\chi)$ and $D_u = \ln(\psi/\chi)$. Contracting the momentum with u_y gives the first equation below; the source law gives the second:

$$\begin{gathered} y\tfrac{dL_u}{dy} + \tfrac{dD_u}{dy} = 2, \qquad e^{L_u} = \tfrac{2y}{\sinh D_u}, \\ \tfrac{dy}{dD_u} + y\coth D_u = 1, \qquad y\sinh D_u = \cosh D_u - 1, \\ y = \tanh(D_u/2), \qquad \psi = 1 + y, \qquad \chi = 1 - y. \end{gathered} \tag{21}$$

The integration constant is fixed by the regular unloaded endpoint $y = D_u = 0$. This earns the exterior before fixing the energy away from that curve. On the curve, the shell momenta are $p^{(y)} = e$, so symmetry leaves only one matrix entry undetermined. The proper local source condition removes it:

$$\begin{gathered} H_{11} = H_{22} = h, \qquad H_{12} = h - 1, \qquad \det H = 2h - 1, \\ e_\alpha^T H^{-1} e_\alpha - 2\alpha = \tfrac{h(\alpha-1)^2}{2h-1} = 0, \qquad e_\alpha = (-\alpha, 1), \\ H = H_0 \text{ on the selected curve}, \qquad (H_0)_{11} = (H_0)_{22} = 0, \quad (H_0)_{12} = -1. \end{gathered} \tag{22}$$

Every loaded point has $\alpha \neq 1$, hence $h = 0$; continuity covers the endpoint. The ambient-response premise is retained on the regular domain used by the field equations, including its response derivatives. Agreement on one background curve alone would not supply that premise.

6. Interacting sources select the remaining energy

Two supported shells probe the energy between distinct response states. Their material law stays fixed: each shell contains a spherical counterrotating population of massive particles. The two rest populations and two angular momenta are four independently prepared physical controls. Appendix C proves a smooth open family of their actual equilibria for each fixed regular candidate at sufficiently weak strength.

The proof uses the candidate's own equations: two asymptotic field values and two mechanical force balances. The source law and energy-charge agreement are subsequently applied to these solutions. No additional prescribed exterior or energy-agreement equation

is used to manufacture their existence. Relative control of the shell separation supplies positive, subluminal sources approaching the exterior curve from one side.

Use the unnormalized radial parameter $\xi = 1/\rho$ in their vacuum annulus, with inner endpoint $\xi_1 = 1/\rho_1$. Define the response-space vector fields $X = (\psi, \chi)$ and $Y = (-\psi, \chi)$. The inner shell's traction and the complete energy identity are

$$\begin{gathered} p^{(\xi)} = Hu_\xi, \qquad K_\xi = u_\xi^T H u_\xi, \qquad X \cdot p^{(\xi)} - \xi_1 K_\xi = 0, \\ E_C - E_s = \tfrac{\kappa}{4} \int (\mathcal{L}_Y H)_{ab} \nabla u^a \cdot \nabla u^b \, d^3x = 0. \end{gathered} \tag{23}$$

The Lie derivative $\mathcal{L}$ measures the change of the quadratic response under its indicated vector field. The second identity follows by varying the same complete energy, retaining its infinity boundary and material stresses. The constant inner cavity contributes no gradients; the selected exterior contributes zero to its integrand. Thus it becomes an annular identity.

The actual controls map openly to outer annular seed data and the inner endpoint. Their local inverse reconstructs each particle population and angular momentum from its jump. Holding a seed fixed while changing the inner endpoint therefore describes actual neighboring sources. Differentiation of (23) along that family gives

$$\begin{gathered} (\mathcal{L}_X H - 2H)(u_\xi, u_\xi) = 0, \qquad (\mathcal{L}_Y H)(u_\xi, u_\xi) = 0, \\ \mathcal{L}_X H = 2H, \qquad \mathcal{L}_Y H = 0. \end{gathered} \tag{24}$$

The second line follows because the actual phase flow supplies an open set of momentum directions at every sampled response point. A quadratic form vanishing on an open set vanishes identically. Appendix C establishes this rank statement from the particle preparations, rather than postulating arbitrary field variations.

The two tensor identities integrate on each connected sampled region. Their constants are fixed by continuity to the already selected exterior curve:

$$\begin{gathered} H_{11} = A_* \tfrac{\chi}{\psi}, \qquad H_{12} = B_*, \qquad H_{22} = C_* \tfrac{\psi}{\chi}, \\ A_* = C_* = 0, \qquad B_* = -1, \qquad F = -\kappa \int \nabla\psi \cdot \nabla(\alpha\psi) \, d^3x. \end{gathered} \tag{25}$$

The field energy has therefore been selected on a genuine connected weak response domain attached to the exterior curve. Since $Z = (X + Y)/2 = (0, \chi)$, the same source identities give $\mathcal{L}_Z H = H$: common clock homogeneity is a consequence there. The existence threshold may depend on the candidate's finite C^2 bounds. Extension to larger loads requires admitted regular source coverage; the weak theorem supplies no universal numerical compactness bound or unrestricted dynamical classification.

7. The selected field energy and its work current

The source tests select the mixed energy on their connected domain:

$$H_{11} = H_{22} = 0, \qquad H_{12} = H_{21} = -1,$$
$$F = -\frac{c^4}{2\pi G}\int \nabla\psi \cdot \nabla(\alpha\psi)\, d^3x. \tag{26}$$

This energy has the reciprocal response symmetry used in the selection proof. Give its infinitesimal parameter a position dependence, eta(x), and vary the two responses oppositely:

$$\delta\psi = -\eta\psi, \qquad \delta\chi = \eta\chi,$$
$$\delta F = \int \mathbf{j} \cdot \nabla\eta\, d^3x, \qquad \mathbf{j} = \kappa(\psi\nabla\chi - \chi\nabla\psi) = \kappa\psi^2\nabla\alpha. \tag{27}$$

The material energy is affected by the same variation. Let σ be its proper stress, with positive isotropic pressure satisfying $\operatorname{tr}\sigma = 3p$. The material and support work gives

$$\delta E_m = 2\int \alpha\psi^6(\varepsilon_m + \operatorname{tr}\sigma)\eta\, d^3x.$$
$$\nabla \cdot \mathbf{j} = 2\alpha\psi^6(\varepsilon_m + \operatorname{tr}\sigma). \tag{28}$$

The second line follows by integrating the field variation by parts and requiring stationarity for an arbitrary compact variation. The physical spatial Laplacian converts the reference current into a local work reading:

$$D^2\alpha = \psi^{-6}\nabla \cdot (\psi^2\nabla\alpha), \qquad \mathbf{J}_W = c^2 D\alpha,$$
$$D \cdot \mathbf{J}_W = \frac{4\pi G}{c^2}\alpha(\varepsilon_m + \operatorname{tr}\sigma). \tag{29}$$

The work current has acceleration units. Its source contains the physical support stresses because those stresses were already included in the same energy. The current is a stationary spatial balance.

A uniform spatial dilation of a closed equilibrium scales the first-gradient field energy by one power of length. The accompanying material work is minus the integrated stress trace. Stationarity therefore gives

$$F = \int \alpha \operatorname{tr}\sigma\, dV, \qquad E_s = \int \alpha(\varepsilon_m + \operatorname{tr}\sigma)\, dV. \tag{30}$$

Integrating (29) over the complete source connects the field current to the same source law:

$$\oint_\Sigma \mathbf{J}_W \cdot d\mathbf{A} = \frac{4\pi G E_s}{c^2} = \frac{Q_s^2}{6V_{\mathrm{ref}}}. \tag{31}$$

Σ is a complete vacuum surface surrounding the source. Its flux is conserved because $D \cdot \mathbf{J}_W = 0$ in vacuum. The finite-domain Q_s remains attached to its actual source domain. It is the combination $Q_s^2/(6V_{\mathrm{ref}})$ that normalizes this conserved work flux.

7.1 Three physical gradient readings

Capacity converts the same clock gradient into three useful readings:

$$\begin{aligned} \mathbf{J}_{\mathrm{cap}} &= -\tfrac{1}{2}D(S^2) = \alpha \mathbf{J}_W, \\ \mathbf{J}_W &= -\tfrac{D(S^2)}{2\alpha}, \qquad \mathbf{a}_{\mathrm{hold}} = \tfrac{\mathbf{J}_W}{\alpha} = c^2 D\ln\alpha. \end{aligned} \tag{32}$$

The last quantity is the outward local acceleration required to hold a probe. The work current is that acceleration expressed with its clock weight. Their different factors matter:

$$D \cdot \mathbf{J}_{\mathrm{cap}} = \alpha D \cdot \mathbf{J}_W + \tfrac{|\mathbf{J}_W|^2}{c^2}. \tag{33}$$

Thus the capacity gradient and the conserved vacuum current have distinct mathematical roles. Their relation follows from the invariant, while conservation follows from the selected energy.

7.2 A field equation written in burden

Put $b_S = S^2 = c^2(1-\alpha^2)$; the subscript distinguishes this burden variable from a gravitational length. Applying the chain rule to (29) gives

$$D^2 b_S + \tfrac{|Db_S|^2}{2(c^2 - b_S)} = -\tfrac{8\pi G}{c^2}\Big(1 - \tfrac{b_S}{c^2}\Big)(\varepsilon_m + \operatorname{tr}\sigma). \tag{34}$$

The nonlinear gradient term follows from the capacity-to-clock conversion. Every term has inverse-time-squared units. The equation uses the positive-clock domain already specified in the construction.

For weak nonrelativistic sources set $\Phi = -S^2/2$ and $\varepsilon_m \simeq \rho_m c^2$. To leading order the physical ruler is Euclidean, and the same equation becomes

$$\nabla^2\Phi = 4\pi G\rho_m, \qquad \mathbf{a} = -\nabla\Phi = \tfrac{1}{2}\nabla S^2. \tag{35}$$

The weak binding energy is consistent with the same complete accounting. The selected field contribution is positive in this decomposition, while the material energy is clock weighted. Together they give

$$\begin{aligned} F &\simeq \tfrac{1}{8\pi G}\int |\nabla\Phi|^2 d^3x = -\tfrac{1}{2}\int \rho_m \Phi \, d^3x, \\ E_s &\simeq E_{\text{proper material}} + \tfrac{1}{2}\int \rho_m \Phi \, d^3x. \end{aligned} \tag{36}$$

For an attractive potential the net binding contribution is negative. The proper material term includes the internal and support energy of the actual preparation.

8. What the source tests distinguish

Two explicit response laws separate the roles of the assumptions. Throughout these comparisons the physical definitions $q_{ij} = \psi^4 \delta_{ij}$ and $\alpha = \chi/\psi$, material coupling, complete energy, and boundary convention are held fixed. The alternatives are static controls within the two-response class; no complete dynamical theory is asserted for them.

An exact source-law survivor

$$H_{\mathrm{alt},\,11} = H_{\mathrm{alt},\,22} = 0, \qquad H_{\mathrm{alt},\,12} = H_{\mathrm{alt},\,21} = -\psi\chi,$$
$$F_{\mathrm{alt}} = -\kappa \int \psi\chi \, \nabla\psi \cdot \nabla\chi \, d^3x. \tag{37}$$

This matrix is smooth and nondegenerate for positive responses and has the selected unloaded normalization. Its reciprocal response symmetry $Y = (-\psi, \chi)$, together with the included support work, gives complete static energy equal to lapse charge. Set $P = \psi^2$ and $W = \chi^2$. Both are harmonic in vacuum, so a spherical exterior has $P = 1 + a/\rho$ and $W = 1 + b/\rho$. With $\mu = (a - b)/2 = GE/c^4$,

$$R = \rho + a, \qquad \alpha^2 = 1 - \tfrac{2\mu}{R}, \qquad B = (1 - \tfrac{a}{R})^{-2}. \tag{38}$$

Here B multiplies dR^2 in the operational spatial interval. The exact source law holds at every admitted actual outer boundary, although B generally differs from the selected radial response. The proper local weak-source coupling instead satisfies $G_{\mathrm{loc}}/G = 1/(\psi\chi)$. The same local source law therefore excludes this alternative on the compared ambient domain. Capacity subtraction alone does not do so. A single hollow-shell cavity has $\psi\chi = 1$ and cannot by itself expose the difference.

Two actual supported hosts can hold the same ruler $\psi = \sqrt{3/2}$ while giving clocks $2/3$ and $1/\sqrt{2}$. Their proper force ratio is $2\sqrt{2}/3$. Increasing the host sizes removes local gradients while retaining that ratio. This also tests the separate complete-process clock principle: $F_{\mathrm{alt}}(\psi, a_{\mathrm{c}}\chi) = a_{\mathrm{c}}^2 F_{\mathrm{alt}}$, whereas material energy has degree one.

A clock-homogeneous family rejected by exact source Q

$$H_{k,\,11} = k\alpha, \qquad H_{k,\,12} = H_{k,\,21} = k - 1, \qquad H_{k,\,22} = k/\alpha,$$
$$d = 1 - 2k > 0, \quad U^2 = 1 - y^2/d, \quad v = -\sqrt{d}\,\operatorname{atanh}(y/\sqrt{d}), \tag{39}$$
$$\psi = Ue^{-v}, \quad \chi = Ue^{v}, \quad y = \tfrac{GE}{2c^4\rho}.$$

The dimensionless constant k labels regular candidates; the displayed curve has $0 < y < \sqrt{d}$. Each has complete static energy equality, degree-one clock homogeneity, the same proper local weak-source coupling, and the same leading perihelion coefficient. Yet its actual shell surface gives

$$\psi_s^2 - \chi_s^2 = 4y_s - \tfrac{16k}{3d} y_s^3 + O(y_s^5). \tag{40}$$

Exact source Q requires the left side to equal $4y_s$ on an open weak source family, selecting $k = 0$. For $k = 0.1$ and $y_s = 0.25$, the ratio of the measured surface Q^2 to its complete-source

prescription is 0.9892755572, although energy and support balance close. Thus exact source composition adds information beyond the leading orbit calculation.

9. Spherical motion and perihelion

Application domain. Take an isolated spherical source whose relevant exterior lies in the connected source-covered domain of the selection result. Applying it to a specified finite source requires that coverage, or the separately conditional continuation of Appendix D. The weak theorem supplies a candidate-dependent attached collar; it gives no candidate-independent compactness threshold guaranteeing the solar application.

9.1 From the selected responses to areal radius

Let ρ be isotropic radius. In the selected exterior, the source energy fixes the length b and the dimensionless load y:

$$\begin{gathered} b = \frac{GE_s}{c^4} = \frac{\mu_Q}{c^2}, \qquad y = \frac{b}{2\rho}, \\ \psi = 1 + y, \qquad \chi = 1 - y, \qquad \alpha = \frac{\chi}{\psi}, \qquad q_{ij} = \psi^4 \delta_{ij}. \end{gathered} \tag{41}$$

The area of a coordinate sphere is $4\pi\rho^2\psi^4$. Its areal radius and radial derivative therefore obey

$$\begin{gathered} r = \rho(1+y)^2, \qquad \frac{dr}{d\rho} = 1 - y^2, \\ A = \alpha^2 = \left(\frac{1-y}{1+y}\right)^2 = 1 - \frac{2b}{r}, \\ B = \psi^4 \left(\frac{d\rho}{dr}\right)^2 = \left(\frac{1+y}{1-y}\right)^2 = A^{-1}. \end{gathered} \tag{42}$$

These are coordinate conversions of the selected clock and ruler responses. They give the operational interval and burden reading

$$\begin{gathered} c^2 d\tau^2 = Ac^2 dT^2 - A^{-1} dr^2 - r^2 d\Omega^2, \\ S^2 = c^2(1 - A) = \frac{2\mu_Q}{r}, \qquad d\Omega^2 = d\theta^2 + \sin^2\theta \, d\varphi^2. \end{gathered} \tag{43}$$

The interval has the Schwarzschild form. Its use here follows the source selection; it is not an additional field equation.

9.2 The retained probe energy gives the orbit

The established common-energy probe coupling is retained. Test-body order means that the probe reads the supplied field without changing its source normalization. For a massive probe with fixed local rest energy $E_0 = mc^2$, its positive Hamiltonian and corresponding clock and momentum readings are

$$E_p = \alpha\sqrt{E_0^2 + c^2 q^{ij} p_i p_j}, \qquad \beta = \frac{d\tau}{dT} = \frac{E_0 A}{E_p}, \\ p_i = \frac{E_0 q_{ij} v^j}{c^2 \beta}, \qquad v^i = \frac{dx^i}{dT}. \tag{44}$$

The Legendre transform gives the action, rather than requiring a separate orbit postulate:

$$L_p = p_i v^i - E_p = -mc^2\beta, \qquad \mathcal{I}_p = -mc^2 \int d\tau. \tag{45}$$

Appendix E records the common-energy assumptions behind (44). Choose the equatorial plane and use dots for d/dT. Time independence and rotational symmetry give

$$\beta^2 = A - \frac{B\dot{r}^2 + r^2\dot{\varphi}^2}{c^2}, \\ \varepsilon = \frac{E_p}{mc^2} = \frac{A}{\beta}, \qquad j = \frac{p_\varphi}{m} = r^2\frac{d\varphi}{d\tau}. \tag{46}$$

Divide capacity by β^2, multiply by A, and substitute these two conserved quantities. The proper-time radial first integral is

$$AB\left(\frac{dr}{d\tau}\right)^2 + A\left(c^2 + \frac{j^2}{r^2}\right) = c^2\varepsilon^2. \tag{47}$$

For $j \neq 0$, set $w = 1/r$ and use primes for $d/d\varphi$. Since $dr/d\tau = -jw'$, the selected $AB = 1$ gives

$$j^2(w')^2 = c^2(\varepsilon^2 - 1) + 2\mu_Q w - j^2 w^2 + \frac{2\mu_Q j^2 w^3}{c^2}. \tag{48}$$

Differentiation yields the orbit equation. The radial Euler equation in Appendix E supplies the same result at turning points, where division by w' would be unavailable:

$$w'' + w = \frac{\mu_Q}{j^2} + \frac{3\mu_Q w^2}{c^2}. \tag{49}$$

The coefficient three is the derivative of the cubic term in (48). That term comes from multiplying the selected clock response by the angular-motion contribution.

Write $p_0 = j^2/\mu_Q$ and $w_0 = (1 + e\cos\varphi)/p_0$. To first order in b/p_0, the perturbation satisfies

$$\delta w'' + \delta w = \frac{3b}{p_0^2}\left(1 + \frac{e^2}{2} + 2e\cos\varphi + \frac{e^2}{2}\cos 2\varphi\right), \\ \delta w_{\text{res}} = \frac{3be}{p_0^2}\varphi\sin\varphi. \tag{50}$$

The resonant term is the first-order expansion of $(e/p_0)\cos[(1-3b/p_0)\varphi]$. Successive perihelia are therefore separated by an angle exceeding 2π by

$$\Delta\varpi = \frac{6\pi\mu_Q}{p_0 c^2} + O\left(\frac{b^2}{p_0^2}\right), \qquad p_0 = a_{\mathrm{orb}}(1-e^2) \text{ at leading order,}$$
$$\Delta\varpi \simeq \frac{6\pi G E_s}{a_{\mathrm{orb}}(1-e^2)c^4} = \frac{3Q_s^2}{16\pi R_0^3 a_{\mathrm{orb}}(1-e^2)c^2}. \tag{51}$$

The last form uses the fixed source-domain law $Q_s^2 = 32\pi^2\mu_Q R_0^3$. The source radius R_0 remains distinct from the orbital semimajor axis a_{orb} and the variable evaluation radius r.

9.3 Mercury as a numerical application

For a reproducible leading monopole calculation, use the J2000 coefficients in JPL's approximate-element Table 1, with the solar parameter and unit conversions from JPL's astrodynamic parameters. These fitted approximate elements are suitable for this numerical illustration; they are not a precision ephemeris fit.

$$\mu_\odot = 1.3271244004127942 \times 10^{20}\ \mathrm{m^3\,s^{-2}},$$
$$a_{\mathrm{orb}} = 0.38709927\ \mathrm{au}, \qquad e = 0.20563593, \tag{52}$$
$$c = 299792458\ \mathrm{m\,s^{-1}}, \qquad 1\ \mathrm{au} = 149597870700\ \mathrm{m}.$$

Substitution gives the orbital scale, small expansion parameter, and advance per radial cycle:

$$p_0 \simeq 5.546046913 \times 10^{10}\ \mathrm{m}, \qquad \frac{b}{p_0} = 2.662482056 \times 10^{-8},$$
$$\Delta\varpi \simeq 5.018660440 \times 10^{-7}\ \mathrm{rad} = 0.1035173023\ \mathrm{arcsec}. \tag{53}$$

The NASA Mercury Fact Sheet gives the sidereal period as 87.969 days. Using 36,525 days per Julian century gives

$$\Delta\varpi_{\mathrm{century}} \simeq 0.1035173023\ \frac{36525}{87.969}\ \mathrm{arcsec} = 42.9807\ \mathrm{arcsec}. \tag{54}$$

Rounded to the appropriate displayed accuracy, this is 42.98 arcseconds per century, the leading relativistic benchmark quoted by Will, Physical Review Letters 120, 191101 (2018). It is the isolated spherical contribution. A full Mercury ephemeris also includes planetary perturbations, solar multipoles and rotation, and smaller relativistic terms. No measured perihelion residual enters the source selection or fixes the coefficient in (49).

9.4 Radial light and the held-clock reading

Take the massless branch of the same Hamiltonian, $E_0 = 0$, with radial momentum only. It is the radial null branch of (43). Its canonical coordinate speed is

$$E_\gamma = c\sqrt{\frac{A}{B}}\,|p_r|, \qquad \left|\frac{dr}{dT}\right| = c\sqrt{\frac{A}{B}} = cA. \tag{55}$$

A held observer reads radial length $d\ell = \sqrt{B}\,|dr|$ and proper clock increment $d\tau_{\text{held}} = \sqrt{A}\,dT$. Those operational readings give

$$\frac{d\ell}{d\tau_{\text{held}}} = \sqrt{\frac{B}{A}}\left|\frac{dr}{dT}\right| = c, \qquad dT = \frac{|dr|}{cA}. \tag{56}$$

Thus the same selected field preserves the local light speed while increasing the reference travel time. For outward radial propagation between $r_2 > r_1 > 2b$, integration gives

$$\Delta T = \frac{r_2 - r_1}{c} + \frac{2b}{c}\log\left(\frac{r_2 - 2b}{r_1 - 2b}\right). \tag{57}$$

The logarithmic term is the excess over the flat radial reference time for these endpoints. This radial example does not require a further light-propagation parameter.

10. Source, capacity and motion in one calculation

The physical requirement is straightforward: the same source law must describe an added source in local field readings and a complete source assembled with other matter. The field, material and supporting stresses participate in one energy account. These requirements restrict how the clock and spatial responses can depend on each other.

The derivation begins with unknown energy coefficients. Actual source balance determines the exterior, and interacting source preparations determine the energy on the domain they explore. The selected reciprocal response supplies the work current and its normalization in Q. The established probe energy then carries that field response into motion, radial light propagation and perihelion.

The field energy has an upstream source-based justification. Its spherical motion follows through the established probe coupling.

Appendix A. Weak one-shell existence and normalization

For the material constructions set $c = G = 1$, so energy has length units. Use $\xi = 1/\rho$, $p^{(\xi)} = Hu_\xi$, and $K_\xi = (p^{(\xi)})^T H^{-1} p^{(\xi)}$. A particle shell of rest energy M and angular length ℓ has the following proper energy, integrated pressure, outward jump and independent traction:

$$\begin{gathered} U = M\sqrt{1+\ell^2/r^2}, \qquad P = \frac{U\ell^2}{2(r^2+\ell^2)}, \qquad r = \rho\psi^2, \\ \Delta p^{(\xi)} = \Big(\frac{\alpha(U+4P)}{2\psi}, -\frac{\alpha U}{2\chi}\Big), \qquad \alpha P = \frac{\Delta K_\xi}{2\rho}. \end{gathered} \tag{58}$$

The outward jump means exterior minus interior. For an empty cavity the interior momentum vanishes. Let $v = H_\infty^{-1} e$ and retain the ordinary attractive calibration $e^T v = 2$, also supplied by the unloaded local source law. Prepare $M = \varepsilon m$, $\ell^2 = \varepsilon j$, and $u_c = (1,1) + \varepsilon a$, with fixed $m, j > 0$. The two asymptotic residuals divided by ε and the traction divided by ε^2 tend to

$$\begin{gathered} G_{\mathrm{AF}} = a - \frac{m}{2\rho} v, \qquad G_{\mathrm{force}} = \frac{mj}{2\rho^2} - \frac{m^2}{4\rho}, \\ \rho_0 = \frac{2j}{m}, \qquad a_0 = \frac{m}{2\rho_0} v, \qquad \det DG = -\frac{m^2}{4\rho_0^2} \neq 0. \end{gathered} \tag{59}$$

In the vacuum flow write $u = (1,1) + \varepsilon a(\xi)$ and $p^{(\xi)} = \varepsilon \bar{p}(\xi)$. Its rescaled equations are

$$\begin{gathered} a_\xi = H^{-1}\bar{p}, \qquad (\bar{p}_b)_\xi = \frac{\varepsilon}{2} a_\xi^T (\partial_b H) a_\xi, \\ \alpha_c = 1 - \frac{\varepsilon m}{\rho_0} + O(\varepsilon^2), \qquad \frac{2P}{U} = \frac{\varepsilon j}{\rho_0^2} + O(\varepsilon^2). \end{gathered} \tag{60}$$

Finite C^2 bounds give convergence with one derivative on compact rescaled sets. The implicit-function theorem therefore supplies actual positive-energy, subluminal weak shells and an open endpoint interval. Only the three own field/force equations enter this construction. An extendible regular endpoint chart permits trial variations; the resulting physical solutions have $0 < \alpha < 1$.

After enclosure has earned $E_s = E_C$, return to $y = \mu/(2\rho)$. On the same exterior $p^{(\xi)} = (\mu/2) p^{(y)}$ and $K_\xi = (\mu^2/4) K_y$. A hollow shell at $y = \delta$ obeys, for arbitrary H,

$$\begin{gathered} \frac{F_{\mathrm{ext}}}{E_C} = \frac{K_y \delta}{2}, \qquad F_{\mathrm{ext}} = 2P_g, \\ \psi p_\psi^{(y)} = 1 + \frac{K_y\delta}{2}, \qquad \chi p_\chi^{(y)} = -1 + \frac{K_y\delta}{2}. \end{gathered} \tag{61}$$

Regularity first gives $p_\infty^{(y)} \to e$ and $K_y \to 2$. Keep the source-dependent initial correction $p_\infty^{(y)} = e + \delta b + o(\delta)$, and write $v = (\gamma, \gamma - 2)$. Let $f_a = \frac{1}{2} v^T (\partial_a H)_\infty v$. Expansion of (61) and exact charge normalization give

$$\begin{gathered} b + f + (\gamma, 2-\gamma) = (1,1), \qquad v \cdot b = 0, \qquad v \cdot f = 2 - 2\gamma, \\ e \cdot u_{yy}(0) = -v \cdot f = 2\gamma - 2. \end{gathered} \tag{62}$$

The geodesic contraction in the second line follows directly by contracting the first-kind connection with v three times. The moving initial momentum contributes $2v \cdot b = 0$, rather than being set to zero. At the actual material endpoint,

$$\begin{gathered} \psi_s^2 - \chi_s^2 = 4\delta + 6(\gamma - 1)\delta^2 + o(\delta^2) = 4\delta, \\ \gamma = 1, \qquad v = e, \qquad H_\infty e = e. \end{gathered} \tag{63}$$

The rightmost equality is exact source Q. This establishes the normalized tangent without assuming clock homogeneity or a common finite-source exterior. The support remains positive: $P/U = \delta/2 + o(\delta)$.

Appendix B. Source enclosure and the common exterior

Restore c and G for the enclosure calculation. Before $H_\infty e = e$ is known, an asymptotically weak enclosing shell adds the field amplitudes shown below. The weak calibration already fixes its lapse-charge increment:

$$\begin{gathered} \delta(a_\psi, a_\chi) = \tfrac{m_B}{4\pi\kappa} H_\infty^{-1} e + O(L^{-1}), \qquad \delta E_C = m_B + O(L^{-1}), \\ \alpha_{AB}(L)^2 = 1 - \tfrac{2GE_{C,AB}}{c^4 L} + o(L^{-1}) = 1 - \tfrac{2GE_{AB}}{c^4 L}. \end{gathered} \tag{64}$$

Self and interaction field energies are $O(L^{-1})$; finite inner susceptibility gives the same order for inner readjustment. Included particle support has vanishing relative pressure. These are the regular distant-cluster conditions of the source class. Multiplying the last equality by L and taking the limit proves (17).

To fix the remaining asymptotic ruler charge, express the local-response identity throughout its ambient domain using $I = \psi\chi$ and $v_c = \tfrac{1}{2}\ln\alpha$. The target quadratic form g then has $g^{v_c v_c} = 1/(2I)$, equivalently

$$\begin{gathered} g = \mathcal{A}\, dI^2 + 2\mathcal{B}\, dI\, dv_c + \left(2I + \tfrac{\mathcal{B}^2}{\mathcal{A}}\right) dv_c^2, \qquad \mathcal{A} \neq 0, \\ \mathcal{B}_\infty = 0, \qquad \mathcal{C}_\infty = \left(\tfrac{1}{2}\partial_I \mathcal{B} - \tfrac{1}{4}\partial_{v_c}\mathcal{A}\right)_\infty. \end{gathered} \tag{65}$$

For a normalized exterior, $I_y(0) = j = 2\mathcal{D}/\mu$ and $(v_c)_y(0) = -1$. The geodesic equation of (65) gives the following Taylor expansion of the actual source expression:

$$\begin{gathered} \mathcal{F}_Q = -\tfrac{I}{2}\sinh(2v_c), \qquad \mathcal{F}_Q(y) = y + \tfrac{1}{2}(j + \mathcal{C}_\infty j^2)y^2 + o(y^2), \\ \mathcal{F}_Q(y_s) = y_s \quad \text{at each actual outer boundary.} \end{gathered} \tag{66}$$

Once Appendix A is applied, distant weak matching gives $\mathcal{D}_{AB} = \mathcal{D}_A + o(1)$ and $\mu_{AB} = \mu_A + Gm_B/c^4 + o(1)$. Apply (66) at the aggregate's own outer shell and divide by y_s^2. Smoothness gives a uniform remainder for the bounded normalized initial data. The limit yields (19). Applying enclosure also to these aggregates respects the order in which energy-charge agreement was earned.

The resulting common initial data and actual hollow endpoint interval justify every differentiation in (21). The regular solution follows either by its integrating factor $\sinh D_u$ or by geodesic uniqueness after the endpoint is fixed. Its areal readings are

$$r=\rho(1+y)^2, \qquad \alpha=\frac{1-y}{1+y}, \qquad \frac{dr}{d\rho}=1-y^2,$$
$$A=\alpha^2=1-\frac{2\mu}{r}, \qquad B=\frac{\psi^4}{(dr/d\rho)^2}=\frac{1}{A}. \tag{67}$$

These statements hold on the regular exterior interval reached by the source argument. They use the exact law on actual material boundaries followed by solution uniqueness, preserving the distinction between a source domain and a vacuum reading.

Appendix C. Actual two-shell sources and local functional selection

Use the geometric units and shell law of Appendix A. Let $0<\rho_1<\rho_2$, with a constant inner cavity, one vacuum annulus and the outer vacuum. Prepare $M_i=\varepsilon m_i$ and $\ell_i^2=\varepsilon j_i$. Write the unknown cavity fields as $\psi_c=1+\varepsilon\Phi+\varepsilon^2 Z_c$ and $\chi_c=1-\varepsilon\Phi+\varepsilon^2 Z_c$. Divide the antisymmetric and symmetric asymptotic residuals by ε and ε^2, respectively, and both tractions by ε^2. Their limits are

$$G_\Phi=\Phi-\frac{m_1}{2\rho_1}-\frac{m_2}{2\rho_2}, \qquad \Phi_2=\Phi-\frac{m_1}{2\rho_1}+\frac{m_1}{2\rho_2},$$
$$G_Z=Z_c+\frac{1}{2}\Big(-\frac{m_1\Phi}{\rho_1}+\frac{m_1 j_1}{\rho_1^3}-\frac{m_2\Phi_2}{\rho_2}+\frac{m_2 j_2}{\rho_2^3}\Big), \tag{68}$$
$$G_1=\frac{m_1 j_1}{2\rho_1^2}-\frac{m_1^2}{4\rho_1}, \qquad G_2=\frac{m_2 j_2}{2\rho_2^2}-\frac{m_2(2m_1+m_2)}{4\rho_2}.$$

Their positive root and Jacobian, in unknown order $(\Phi, Z_c, \rho_1, \rho_2)$, are explicit:

$$\rho_1=\frac{2j_1}{m_1}, \qquad \rho_2=\frac{j_2}{m_1+m_2/2}, \qquad \Phi=\frac{m_1}{2\rho_1}+\frac{m_2}{2\rho_2},$$
$$Z_c=\frac{m_1 m_2}{4\rho_2}\Big(\frac{1}{\rho_1}-\frac{1}{\rho_2}\Big), \qquad \det DG=\frac{m_1^2 m_2(2m_1+m_2)}{16\rho_1^2\rho_2^2}>0. \tag{69}$$

To control a general candidate, use normal and tangent response coordinates $n=(\psi+\chi-2)/2$ and $w=(\psi-\chi)/2$. Equality to H_0 on $n=0$, together with its own selected-line geodesicity, implies

$$g_{nn}=-2+O(n), \qquad g_{nw}=O(n), \qquad g_{ww}=2+O(n^2),$$
$$n=\varepsilon^2\zeta, \quad w=\varepsilon\phi, \quad p_n=\varepsilon^2\eta, \quad p_w=\varepsilon\Pi, \tag{70}$$
$$\zeta_\xi=-\eta/2+O(\varepsilon), \quad \phi_\xi=\Pi/2+o(1), \quad \eta_\xi=O(\varepsilon), \quad \Pi_\xi=o(1).$$

The potentially leading normal force contains $\partial_n g_{ww}=O(n)$; mixed normal derivatives carry an extra ε. The estimates hold with one derivative under finite C^2 bounds. Thus the normalized own boundary problem converges in C^1 to (68), and its nonzero determinant

supplies independent actual particle controls. Neither vanishing normal first derivatives nor small constitutive deformation is assumed.

Attachment to the line requires relative separation control. Parametrize exact H_0 particle data by a gap $g \geq 0$, keeping masses positive and radii bounded away from zero. At the ordered zero-gap reference, the annulus has zero length; the candidate and H_0 have exactly the same root because their line values and exterior geodesic coincide. The determinant remains nonzero there. Uniform continuation and differentiation in g give

$$\begin{gathered}(\rho_2-\rho_1)_H=(\rho_2-\rho_1)_{H_0}[1+O(\varepsilon)],\\ \frac{n}{\varepsilon^2}=\frac{m_1m_2}{4\rho_2}\left(\xi-\frac{1}{\rho_2}\right)+o(1),\qquad \frac{w}{\varepsilon}=\frac{m_2}{2\rho_2}+\frac{m_1\xi}{2}+o(1).\end{gathered} \tag{71}$$

For sufficiently small fixed strength and positive arbitrarily small gap, the physical shells remain separated, their energies positive and their support speeds subluminal. The reference with zero separation is only a limiting point of the proof. These sources reach a connected side of the line at nonzero weak load.

For every passing actual source its outer exterior is already selected. At $\xi_2=1/\rho_2$, put $y=\mu\xi_2/2$, $u_s=(1+y,1-y)$, and $p_{\mathrm{out}}^{(\xi)}=(\mu/2)e$. The annular momentum p below is $p^{(\xi)}$. Combining the outer jump with its traction leaves exactly three seeds:

$$\begin{gathered}\psi_s p_\psi+\chi_s p_\chi+2\xi_2 p_\psi p_\chi=0,\qquad p_\psi=-\frac{\chi_s p_\chi}{\psi_s+2\xi_2 p_\chi},\\ (\mu,\xi_2,p_\chi,\xi_1),\qquad \xi_1=1/\rho_1.\end{gathered} \tag{72}$$

These four seed/endpoint variables have an actual open image. To see this, integrate the candidate annulus, read U_i,P_i from each jump, and reconstruct its controls:

$$\begin{gathered}M_i=\sqrt{U_i(U_i-2P_i)},\qquad \ell_i^2=\frac{2P_ir_i^2}{U_i-2P_i},\qquad 0<2P_i<U_i,\\ \mathcal{G}_H(\mathcal{S}_H(z))=z,\qquad D\mathcal{G}_H\,D\mathcal{S}_H=\mathrm{Id}.\end{gathered} \tag{73}$$

Here $z=(M_1,\ell_1^2,M_2,\ell_2^2)$ and $\mathcal{S}_H$ is the actual source map earned above. Its smooth left inverse proves full rank. Hence an endpoint variation at fixed seed is an actual source variation. At the selected section $\Sigma=\psi+\chi-2=0$, positive inner and outer populations give

$$-\mu/2<p_\chi<0,\qquad \frac{d\Sigma}{d\xi}=-\frac{2\xi_2p_\chi(\mu/2+p_\chi)}{\psi_s+2\xi_2p_\chi}>0. \tag{74}$$

Three seed coordinates and elapsed affine parameter therefore form a local flow box in four-dimensional (u,p) space. Regular geodesic flow preserves its rank. Each sampled response point has an open momentum fiber, which is the required set of physical directions.

The complete-energy identity can be derived before localizing it. Let σ be the proper material stress including support, and put $\mathcal{T}=\int\alpha\mathrm{tr}\sigma\,dV$. Spatial dilation gives the closed-source virial $F=\mathcal{T}$. The variation generated by Y changes both clock and physical area; its material term and the retained infinity boundary are

$$\begin{gathered}\delta_Y E_m = 2E_m + 2\mathcal{T} = 2E_s,\\ B_Y = 2E_s + \tfrac{\kappa}{2}\int(\mathcal{L}_Y H)_{ab}\nabla u^a\cdot\nabla u^b\,d^3x, \qquad B_Y = 4\pi\kappa\mu = 2E_C.\end{gathered} \tag{75}$$

The boundary value uses the earned $H_\infty = H_0$ and the actual asymptotic charge. This proves (23) with material and support terms retained.

Finally, the inner jump and traction give $X\cdot p-\xi_1 K_\xi = 0$. Differentiate at fixed seed. For the energy identity, radial integration converts the annular term to $4\pi\int_{\xi_2}^{\xi_1}(\mathcal{L}_Y H)(u_\xi,u_\xi)d\xi$. The selected exterior has zero integrand by line agreement and geodesicity. Therefore

$$\begin{gathered}\tfrac{d}{d\xi_1}(X\cdot p-\xi_1 K_\xi) = \tfrac{1}{2}(\mathcal{L}_X H - 2H)(u_\xi,u_\xi) = 0,\\ \tfrac{d}{d\xi_1}\int_{\xi_2}^{\xi_1}(\mathcal{L}_Y H)(u_\xi,u_\xi)\,d\xi = (\mathcal{L}_Y H)(u_\xi,u_\xi) = 0.\end{gathered} \tag{76}$$

Open momentum fibers give (24) pointwise. Their component equations first make H degree zero under X; the Y equations then give precisely the three-constant form in (25). Connected attachment fixes those constants. This proves exact functional equality on the attained weak domain, including smooth terms invisible in every Taylor coefficient along the exterior line.

Appendix D. Conditional continuation along the exterior line

The weak source theorem is the main selection result. The following extension is separate and conditional: on a connected regular C^2 response domain, suppose the same complete-source law, proper local weak-source response, iterated enclosing-source closure, and classical particle-shell preparations apply to all the positive, timelike sources constructed below. Then the attached selected collar continues along every compact line interval $0<y\le Y<1/2$. The physical applicability of those source laws is an input to this statement.

D.1 A regular ordered boundary at each finite load

Use $G=c=1$ in this appendix. The selected two-shell family has outer isotropic radius L, inner radius Lx/t, and outer load y. Its cavity fields are

$$\rho_2 = L, \qquad \rho_1 = \tfrac{Lx}{t}, \qquad \psi_c = \tfrac{(1+y)(1+t)}{1+x}, \qquad \chi_c = \tfrac{(1-y)(1-t)}{1-x}. \tag{77}$$

Only $t > x$ describes two distinct shells. The ordered boundary $t = x$ is a smooth extension of the boundary-value equations, used as an implicit-function anchor. With U_i the proper shell energy and P_i the integrated tangential pressure, its particle-speed ratios are

$$\begin{gathered} w_1 = \frac{2P_1}{U_1} = \frac{x}{1-x}, \qquad w_2 = \frac{2P_2}{U_2} = \frac{x+y}{(1-x)(1-y)}, \\ 0 < x < \min\left[y, \tfrac{1}{2}, \frac{1-2y}{2-y}\right], \qquad 0 < y < \tfrac{1}{2}. \end{gathered} \tag{78}$$

This interval is nonempty; it makes both energies positive and both w_i strictly between zero and one. Fix each shell's rest population and specific angular momentum. The direct physical boundary-value problem has four unknowns: logarithms of the cavity fields and isotropic radii. Its equations impose both asymptotic normalizations and both individual normal-force balances.

At $L = 1$ and $t = x$, using unnormalized force rows, its Jacobian determinant is

$$\begin{aligned} \det J &= -\frac{x^2(x-y)(4x^2-7x+1)P(x,y)}{(x-1)^4(x+1)^4}, \\ P(x,y) &= x^3(y^2-7y+4) + x^2(y^3-10y^2+18y-7) \\ &\quad + x(-7y^3+18y^2-10y+1) + 4y^3 - 7y^2 + y. \end{aligned} \tag{79}$$

For fixed positive $y < 1/2$, P cannot vanish identically in x. Its constant coefficient is $y(4y^2 - 7y + 1)$. At the sole zero of that coefficient in this interval, the linear coefficient remains nonzero:

$$y_c = \frac{7-\sqrt{33}}{8}, \qquad [x]P(x, y_c) = \frac{147 - 29\sqrt{33}}{128} \neq 0. \tag{80}$$

Only finitely many splits are excluded. Thus every such load admits a causal split with a nonsingular direct boundary-value Jacobian. Dividing force rows by positive rest populations preserves this rank.

D.2 The continuation does not assume the next exterior

Suppose selected open collars extend up to a first stopping load $y_* < 1/2$. On those collars $H = H_0$ with vanishing first and second target derivatives. Their C^2 continuity fixes the same two jets at the stopping point and along its whole preceding exterior. The candidate's ordered boundary problem therefore has the reference value and first Jacobian there. Linearized exterior geodesic flow requires only H, its first derivatives, and its second derivatives.

Keep the independent asymptotic ruler-charge defect D while constructing nearby solutions. In the reference normalization, the asymptotic coefficients are $\psi = 1 + (\mu + D)/(2\rho) + \cdots$ and $\chi = 1 + (D - \mu)/(2\rho) + \cdots$, with $\mu = b$ when the common exterior is selected. Define four annular seed/end variables

$$\begin{gathered} \mathcal{S} = (\mu, \tau_2, p_{\chi,\mathrm{in}}, \tau_1), \qquad \tau_i = \rho_i^{-1}, \\ \det \frac{\partial \mathcal{S}}{\partial(\log L, x, y, t)} = \frac{2(1+y)}{x(1+x)^2} \neq 0. \end{gathered} \tag{81}$$

The physical inventory map is regular at the chosen split: its determinant has the same potentially vanishing polynomial factors as (79), with nonzero denominators under the strict inequalities. Write the full five-equation problem as $\mathcal{R}(D,S)=0$ and $\mathcal{M}(D,S)=z$, where z comprises the four fixed particle controls and $\mathcal{R}$ is inner empty-cavity traction. At the ordered reference point,

$$\begin{gathered}\mathcal{R}_D=-1, \qquad \mathcal{R}_S=0, \qquad \det \mathcal{M}_S\neq 0,\\ dD=0, \qquad dS=\mathcal{M}_S^{-1}dz.\end{gathered} \tag{82}$$

The unrestricted implicit-function theorem consequently gives an open seed image before applying the source selection to nearby solutions. Its positive-gap half contains distinct positive, timelike shells at outer loads on both sides of y_*. For these actual sources, the already established common-exterior result forces $D=0$. Thus the rank proof has not restricted an unproved exterior charge off shell.

Take the ordered limit from those actual positive-gap sources. The source law and summed shell traction pass continuously to the limit. In particular, $NP_i=(K_{\mathrm{out},\,i}-K_{\mathrm{in},\,i})\tau_i/2$, where $N=\chi/\psi$ and K is the vacuum radial quadratic first integral. At a common limiting radius the intermediate K cancels. The hollow-source endpoint identity and local weak-source law then extend the selected line beyond the proposed stop.

The positive-gap seed domain is open. Its annular flow is transverse to the section $\psi+\chi=2$; the source reconstruction and flowbox argument therefore supply open endpoint states and momenta. The same traction and complete-energy Ward identities used in the weak theorem fix $H=H_0$ on a collar beyond the stop. This excludes a first stopping load inside the stated regular interval.

> This conditional continuation establishes neither a uniform collar thickness nor global target-space reachability or unrestricted dynamical uniqueness. The particle-split interval closes at $y=1/2$. A minimum shell separation or failure of the stated material-source domain would change the premise. The coincident boundary is only a mathematical limit of separately balanced shells; distinct free-particle speeds at one physical radius are not substituted for that family.

Appendix E. Common energy and the regular probe branch

The probe argument uses regular canonical evolution, a fixed positive spatial reading, and additive test-body scaling in a supplied field. An internal action I with rest-energy dependence $E_0=E_{\mathrm{base}}+h(I)$ has phase speed $(\partial E_p/\partial E_0)h'(I)$. Agreement with its clock reading requires $\partial E_p/\partial E_0=\beta$. Scaling both momentum and rest energy gives Euler's identity:

$$\begin{gathered}v^i=\frac{\partial E_p}{\partial p_i}, \qquad \frac{\partial E_p}{\partial E_0}=\beta, \qquad E_p=E_0\beta+p_iv^i,\\ \beta^2=A-\frac{q_{ij}v^iv^j}{c^2}.\end{gathered} \tag{83}$$

The rest-energy derivative compares neighboring internal states at the same momentum. It does not require the probe to change its rest energy during an orbit. Differentiation of Euler's identity at fixed position and rest energy gives

$$\left(p_i + E_0 \frac{\partial \beta}{\partial v^i}\right) \frac{\partial v^i}{\partial p_j} = 0, \qquad \frac{\partial \beta}{\partial v^i} = -\frac{q_{ik} v^k}{c^2 \beta}. \tag{84}$$

Invertibility of the massive-branch momentum Hessian makes the bracket vanish. Euler's identity then yields $E_p = E_0 A/\beta$, the square-root Hamiltonian (44), and the action (45). These coupling assumptions turn the capacity reading into dynamics; the scalar capacity equation alone does not specify a canonical Hamiltonian.

For completeness, variation of this action gives a proper radial equation valid also at an apsis. Substituting $B = A^{-1}$ and $A = 1 - 2\mu_Q/(c^2 r)$ gives

$$\begin{gathered} \frac{d^2 r}{d\tau^2} = -\frac{\mu_Q}{r^2} + \frac{j^2}{r^3} - \frac{3\mu_Q j^2}{c^2 r^4}, \\ \frac{d^2 r}{d\tau^2} = -j^2 w^2 w^{'}. \end{gathered} \tag{85}$$

These identities prove (49) without dividing by a radial velocity and include the circular-orbit limit for $j \neq 0$.

Appendix F. An independent complete-process clock route

A separate sufficient route retains complete-process uniform-clock scaling directly. It uses the same regular two-response quadratic energy class, actual source Q, included material supports, and universal $E = E_{\text{charge}}$. It requires a physical comparison of the same complete internal state at fixed ruler. Uniform ambient clock multiplication by $a_c > 0$ must multiply the full internal Hamiltonian flow by a_c whenever both states belong to the compared positive-capacity domain. This requirement is stronger than ideal-clock rates or a shared propagation speed.

On a constrained state space, common rates constrain the reduced Hamiltonian under a specified physical symplectic identification. To infer a local constitutive identity, retain that identity explicitly, or require actual transfer of two independent supported weak-source responses with the same ruler and relative-clock profiles. The cold and pressured source covectors $(-\alpha, 1)$ and $(-\alpha(1+6t), 1)$, with t the proper pressure-to-energy-density ratio, separate the invertible response matrix. This gives

$$\begin{gathered} J_a^T H(\psi, a\chi) J_a = a H(\psi, \chi), \qquad J_a = \operatorname{diag}(1, a), \\ \mathcal{L}_Z H = H, \qquad Z = (0, \chi), \\ H_{11} = \chi a_f(\psi), \qquad H_{12} = b_f(\psi), \qquad H_{22} = d_f(\psi)/\chi. \end{gathered} \tag{86}$$

Here $\mathcal{L}$ is the response-space Lie derivative; a_f, b_f, d_f are initially arbitrary functions. No simultaneous ruler homogeneity is assumed. For the vacuum parameter $y = GE_{\text{charge}}/(2c^4\rho)$, put $u = (\psi, \chi)$, $p = Hu_y$ and $K = u_y^T H u_y$. The vacuum curve is a response-space geodesic and K is constant. The clock boundary identity gives

$$\begin{aligned} E = B_Z = -E_{\text{charge}} p_\chi(0), \qquad & p_\chi(0) = -1, \\ (p_\psi(0) - 1)(H_\infty^{-1} e)_\psi = 0, \qquad & e = (1, -1). \end{aligned} \tag{87}$$

B_Z denotes the asymptotic variational boundary flux; the subscript on H_∞ denotes the unloaded endpoint. The second identity uses the calibrated weak response $e^T H_\infty^{-1} e = 2$. On the generic branch it fixes $p(0) = e$ and $K = 2$. The exceptional branch $(H_\infty^{-1} e)_\psi = 0$ has weak tangent $(0, -2)$; its regular supported-shell expansion gives $\psi_s^2 - \chi_s^2 = 4y_s - 6y_s^2 + O(y_s^3)$, contradicting actual source Q.

Every passing source consequently has the same normalized exterior curve. Actual hollow-shell jumps and included tangential work give the following identities on the interval sampled by those sources:

$$p_\psi = \frac{1+y}{\psi}, \qquad p_\chi = \frac{y-1}{\chi}, \qquad p \cdot u_y = 2, \qquad \psi^2 - \chi^2 = 4y. \tag{88}$$

Writing $L = \ln(\psi\chi)$ and $D = \ln(\psi/\chi)$ gives $yL_y + D_y = 2$ and $e^L = 2y/\sinh D$. Hence

$$\frac{dy}{dD} + \coth(D) y = 1, \qquad y = \tanh(D/2), \qquad \psi = 1 + y, \quad \chi = 1 - y. \tag{89}$$

The regular unloaded endpoint fixes the integration constant. These equations sample actual shell boundaries; they do not assign a fixed source quantity to arbitrary vacuum screens. Clock homogeneity and $He = e$ on this curve leave

$$\begin{aligned} H = H_0 + \frac{a_f(\psi)}{\chi} v v^T, \qquad & v = (\chi, 2 - \psi), \\ (H_0)_{11} = (H_0)_{22} = 0, \qquad & (H_0)_{12} = (H_0)_{21} = -1. \end{aligned} \tag{90}$$

The remaining square vanishes, with its first variation, on the isolated exterior. A weak supported core in a loaded cavity supplies the direction $g = H^{-1}(-\alpha, 1)$. Put $Y = (-\psi, \chi)$, $W = \mathcal{L}_Y H$, $x = \psi$, and $a = a_f(x)$ at its background $\chi = 2 - x$. With primes denoting d/dx and C an integration constant, the exact energy identity and its leading core contraction are

$$\begin{aligned} E_{\text{charge}} - E &= \tfrac{\kappa}{4} \int (\nabla u)^T W \nabla u \, d^3x, \\ g^T W g &= \frac{4(x-2)(x-1)[-4a^2 + a(x+1) + x(x-1)a']}{x^2[1 + 2a(x-2)]^2}, \\ x(x-1)a' &= a(4a - x - 1), \qquad a \neq 0 \implies a = \frac{x}{2 + C(x-1)^2}. \end{aligned} \tag{91}$$

The core localization uses its own mechanical equilibrium and includes the host response. For candidates passing universal energy equality, their earned common exterior makes its contribution vanish. Every nonzero regular solution tends to $a_f(1) = 1/2$, where H_∞ is singular. Regular nondegeneracy on a connected source-covered interval attached to the unloaded state therefore gives $a_f = 0$ and the mixed functional. Clock scaling extends the

result in χ only where the stated comparisons exist; it does not establish arbitrary ruler-state or strong-field coverage. This sufficient route is independent of the source-based derivation in the main text.

Appendix G. Verification record and physical preparations

The reproducible record is stored under outputs/set-selection-ontology-four-hour-2026-11-01. Its symbolic calculations check identities; numerical equilibria check concrete source preparations. Neither is an astronomical observation or a substitute for the stated universal source-domain premises.

The main preparation checks are matter-route/check_upstream_one_shell_ift.py, with 16 identities and three independently solved nonselected weak shells, and matter-route/check_weak_particle_source_ift.py, with 19 identities. Their recorded normalized boundary-value Jacobians are

$$-\frac{m^2}{4r^2} \neq 0, \qquad \frac{m_1^2 m_2(2m_1+m_2)}{16 r_1^2 r_2^2} > 0. \tag{92}$$

Here $m, m_i > 0$ are the fixed coefficients of the weak rest-population scaling and $r, r_i > 0$ its limiting isotropic radii. The companion weak-particle-candidate-control.json records eight candidate-owned two-shell solutions, maximum normalized residual below 1.5×10^{-13}, positive ordered gaps, and subluminal particle speeds. Doubling the radial integration steps changes the gap by about 1.1×10^{-16}. No energy-equality or source-Q residual is imposed as a solver condition.

The clock route is checked by countermodels/check_clock_homogeneous_static_selector.py, whose JSON records 19 identities. The alternative in Section 8 has 20 identities in independent-referee/check_source_native_algebra.py and three fixed-material equilibria in independent-source-native-counterexample-results.json: energy gaps are below 8×10^{-13} and source-Q^2 ratio errors below 1.8×10^{-12}. The H_k record, root-audit/exact-source-clock-energy-family-results.json, contains 23 identities and 12 supported shells using one fixed surface equation of state $p_s = K_{\mathrm{eos}} \sigma^2$.

> All source energies include material, field, binding and physical support, with the field integrated through its exterior and one fixed asymptotic convention. Thin-wall limits retain material surface energy and stress. Distinct particle shells obey their individual normal-force equations; coincident-shell limits serve only as mathematical continuation anchors. Positive energy, subluminal support and nonsingular source Jacobians establish the preparations used here, not dynamical stability, global target coverage, or a classification beyond the regular conformal two-response static sector.

References and calculation record

E. Echeverria, *Space Emanation Theory - Integrated Rough Draft 6* (2026). Capacity, complete source energy and the retained common-energy probe coupling. Public manuscript.

The source-field algebra, particle-source equilibrium calculations and numerical controls accompanying this paper are indexed in its calculation record. Appendix G identifies the checks and their mathematical role. Astronomical inputs and the comparison for Mercury are cited at the calculation in Section 9.